

Fishery policy exploration in the Pearl River Estuary based on an Ecosim model

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ABSTRACT

The Pearl River Estuary (PRE) is a tropical estuary that was an important fishery ground in the South China Sea, where many other fisheries have also experienced substantial declines in productivity. In this study, we develop an ecological model constructed by Ecopath with Ecosim (EwE) software to explore the consequences of different fishing policies on the management of the PRE ecosystem. Scenarios are examined that maximise the net profits from the fishing activities, the jobs provided by the fisheries and the measures of ecosystem status. Then the trade-offs due to the ecosystem-scale optimisation of fishery management policies are analysed. The results indicate that optimising short-term economic and social (employment) goals in the PRE creates a depletion of the top predators and losses of biodiversity. Alternatively, focusing on the structure of the ecosystem leads to a reduction in fishing effort for almost all fleets and a corresponding increase in the biomass of most fishes at the expense of reduced profits and jobs provided by the fisheries. An ideal policy compromise is found when equal weight placed on profits, jobs and ecosystem structure, which leads to an overall increase in profits, jobs/catch value, ecosystem maturity and biodiversity. The effects of changing weights placed on profits, jobs/catch value and ecosystem maturity on ecosystem indices are examined and the parameters of vulnerabilities and discount rates are also analysed in this paper.

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1. Introduction

Over the past ten years, the marine capture from the Pearl River Estuary (PRE) has reached a critical stage and fish stocks are severely depleted. The ecosystem becomes increasingly vulnerable with a decrease in the proportion of biomass that can be attributed to high trophic level organisms (Duan et al., 2009b). The Chinese government has begun to pay more attention to ecological protection at the national level. Current fishing policies include a fishing licence system, closed zones/seasons and a summer moratorium, a “zero-growth” policy, a fishing buyback program and fighting against illegal, unreported and unregulated (IUU) fishing (Yu and Yu, 2008). Although many of these comprehensive regulations have been implemented, the ecosystem is still in severe decline.

Cheung and Pitcher (2006) suggested that the Northern South China Sea ecosystem could be restored by reducing fishing effort at an annual rate of 5% for 30 years. The buyback of fishing vessels and licenses could be key policies that reduce fishing capacity and restructure fishing vessels to achieve ecological conservation and management objectives. Buyback is an economic mechanism that would internalise costs through a “fishermen paid” scheme that can

improve effectiveness. This study applies a widely used ecological model Ecopath with Ecosim (EwE) to evaluate the effects of fishery policies on the PRE ecosystem. EwE enables the examination of the ecosystem scale effects of harvest rates from multispecies communities. The optimisation results reveal the trades-off between fisheries and ecological conservation objectives. Several studies had suggested that multispecies models can be a useful tool for fisheries management (Pauly et al., 1998; Christensen and Walters, 2004a,b). Fishing policy simulations can help reconstruct the fishing communities to promote the objectives of economic, social and/or ecological criteria.

This study examines policies for managing of the PRE fishery under several scenarios, where economic, employment and ecological criteria are weighted differently. The model also examines the ecosystem structure and provides an explicit and quantitative framework to analyse the impact of fishing policy on the proportion of biomass of organisms from different trophic levels in the ecosystem, as well as the whole ecosystem.

2. Materials and methods

2.1. The case of the Pearl River Estuary

The PRE ecosystem has a depth of less than 60 m and ranges from 112.5°E to 114.5°E (Fig. 1). It falls largely within the Pearl

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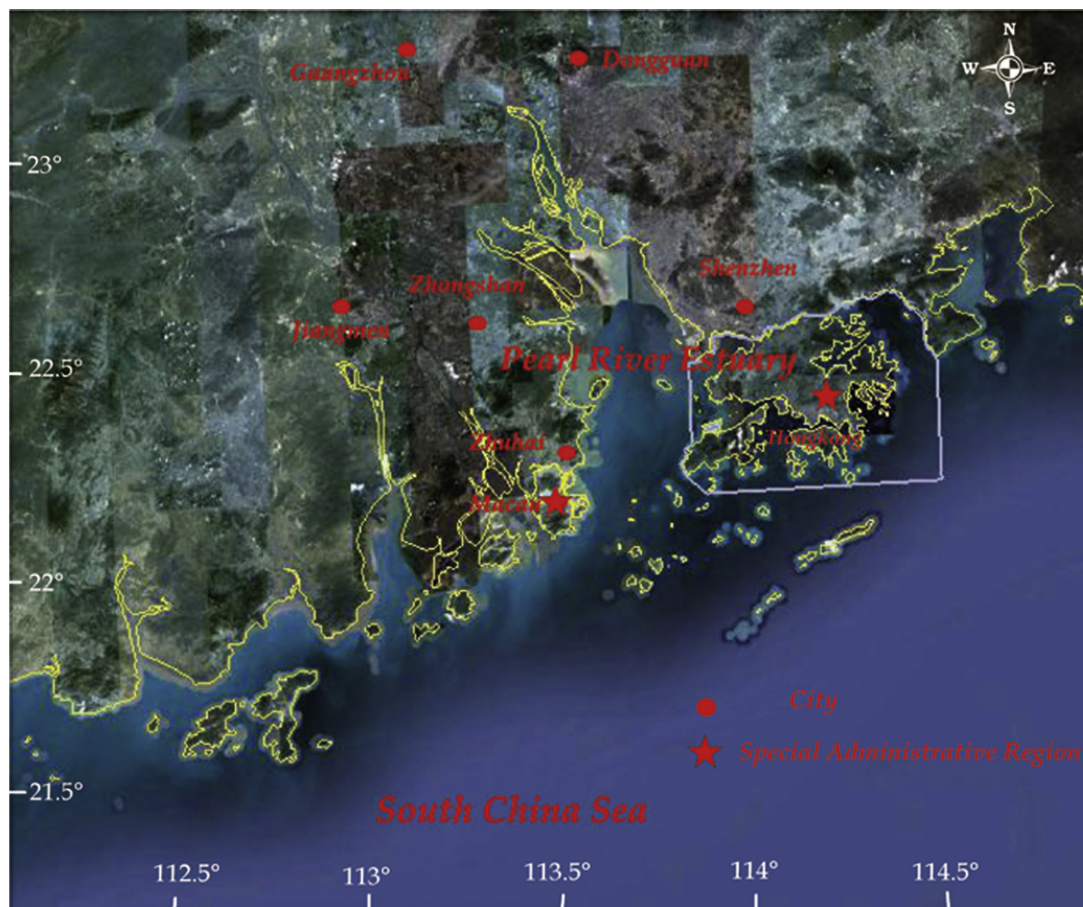


Fig. 1. Geographical scope of this study.

River Delta (PRD) Economic Zone within an area of approximately 72,490 km². The Pearl River consists of three major tributaries, the West River, the North River and the East River that converge at the PRD estuarine region and form a complex river network that is discharged into the South China Sea (Leung et al., 2007). Because of the unique hydrological features of the PRE, fish communities in the Pearl River are complex. Further, diverse ecological mechanisms, including predation, generate a very composite tropical marine ecosystem (Li et al., 2000). However, these fisheries have experienced substantial declines in productivity.

In this paper, we develop EwE model for the Pearl River Estuary (Duan, 2009; Duan et al., 2009a). Some domain species are added in this case study because they are commercially exploited species (see Appendix A for the parameters of the new Ecopath model). The improved version of Ecopath model for the PRE ecosystem consists of 34 functional groups, including marine mammals, seabirds, turtles, Sharks, fishes such as melon seed (*Psenopsis anomala*), pomfret (*Stromateids*), bensasi goatfish (*Upeneus bensasi*), chub mackerel (*Pneumatophorus japonicus*), silver croaker (*Argyrosomus argentatus*), lionhead croaker (*Collichthys lucidus*), greater lizardfish (*Saurida tumbil*), Japanese jack mackerel (*Trachurus japonicus*), threadfin bream (*Nemipterus virgatus*), bigeyes (*Priacanthidae*), Japanese scad (*Decapterus maruadsi*), cutlassfishes (*Trichiuridae*), invertebrates such as shrimp, crabs, zooplankton, jellyfish, polychaeta, mollusks, echinoderms, benthic crustaceans and phytoplankton as primary producers.

Fishery resources are mainly exploited by six fishing methods, including stern and pair trawling, shrimp trawling, gillnet fishery, hook and line, purse seine and other miscellaneous methods. The composition of the catches based on different fishing methods is

presented in Appendix B. The effects of different fishing policies on economic, social and ecosystem structure are analysed in this paper.

2.2. Theoretical approach

The method and theory of Ecopath and Ecosim modelling are detailed in the EwE user guide (Christensen et al., 2005). In short, the Ecopath model creates a static mass-balanced snapshot of the ecosystem during a particular time period. The mass-balance equation assumes that the energy input to any group must equal the energy output from this group, which can be expressed as:

$$\begin{aligned} \text{Production } (P_i) = & \text{catches } (Y_i) + \text{predation mortality } (M_{1i}) \\ & + \text{biomass accumulation } (BA_i) \\ & + \text{net migration } (E_i) + \text{other mortality } (M_{2i}) \end{aligned} \quad (1)$$

The function (1) trophic mass-balance models also can be defined by the following equation:

$$\left(\frac{P}{B}\right)_i \times B_i \times EE_i = \sum_j^n B_j \times \left(\frac{Q}{B}\right)_j \times DC_{ji} + Y_i + E_i + BA_i \quad (2)$$

where i is the prey group, j is the predator group, $(P/B)_i$ is the production/biomass ratio, B_i is the biomass of prey i , EE_i is the ecotrophic efficiency, $(Q/B)_j$ is the consumption/biomass ratio for predator group j , DC_{ji} is the proportion of predator j 's diet

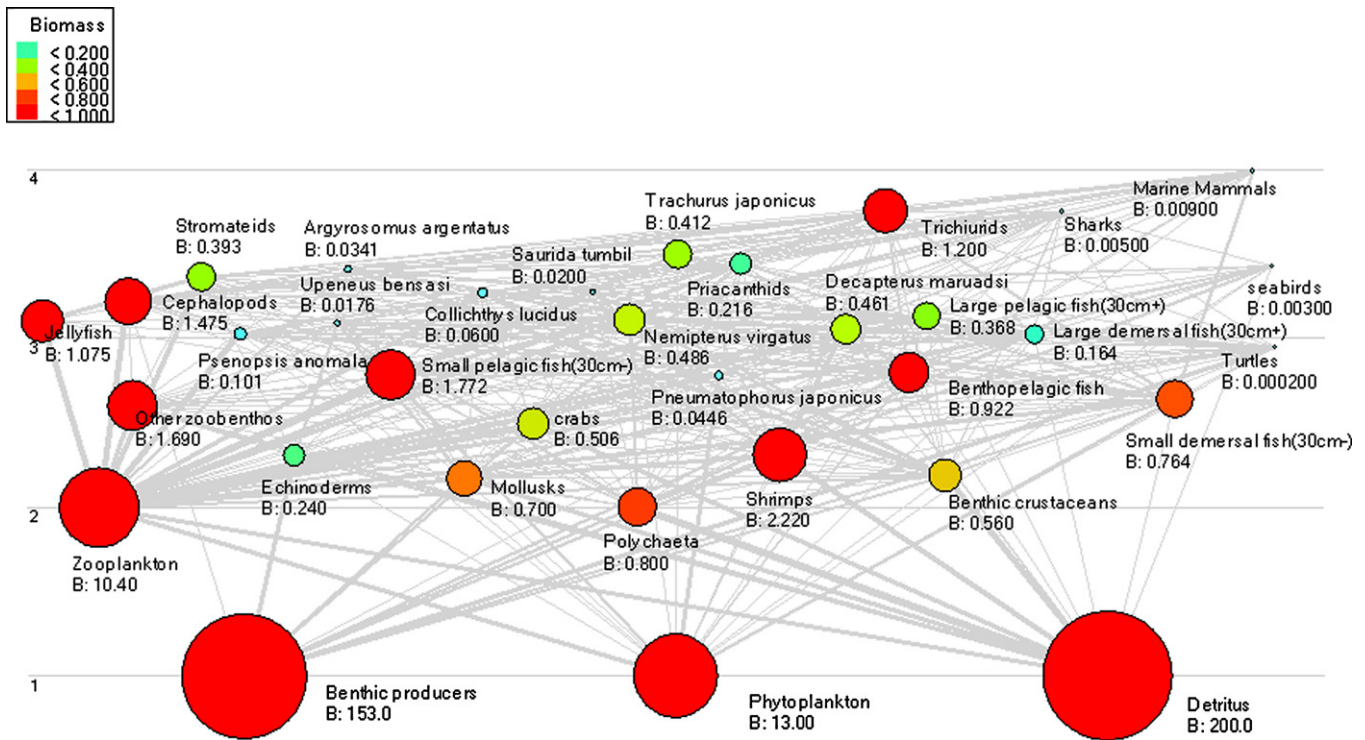


Fig. 2. Flow chart of trophic interaction in the coastal ecosystem of the PRE. (All flows are expressed in $\text{t km}^{-2} \text{ year}^{-1}$.)

contributed by prey i , Y_i is the total catch, E_i is the net migration of prey i , and BA_i is the biomass accumulation of group i .

Ecosim is a time-dynamic model that simulates changes in the ecosystems described under Ecopath and is governed by the basic equations derived from the Ecopath master equation (Eq. (1)):

$$\frac{dB_i}{dt} = g_i \sum_j Q_{ji} - \sum_j Q_{ij} + I_i - (M_i + F_i + E_i) \times B_i \quad (3)$$

where dB_i/dt is the biomass changes of the group, g_i is the net growth efficiency (production/consumption), I_i is the immigration rate, M_i is the nature mortality rate (non-predation mortality), F_i is the fishing mortality rate, and E_i is the emigration rate.

Ecosim uses the Davidson–Fletcher–Powell (DFP) nonlinear optimisation procedure by changing the relative fishing rates (Christensen et al., 2005). Because nonlinear optimisation may provide unrealistically extreme answers due to inappropriate objective functions, all optimisations are begun with random fleet efforts and repeated at least 10 times with an array of weighting factors (Christensen and Walters, 2004b).

Several scenarios were simulated to explore the optimal fishing policy in the PRE: (1) Optimising the economic benefit (\$), which is calculated as the value of the catch (catch-price by species, unit: Euro), is the first scenario. The costs of operating of the different fleets, which was collected from field survey in 2009, are shown in Table 1. (2) Optimising social benefits (J), which is related to the jobs/catch value ratio, and the social index (jobs/catch value, Euro) provided by the 2004 Fisheries Yearbook is the second scenarios, and has the following values: pair and stern trawling, 1.5; shrimp trawling, 2; purse seine, 3; hook and line, 6; gill net, 5; other miscellaneous methods, 2. (3) Optimisation ecosystem status (E) is the third scenarios and was based on one of Odum's measures of ecosystem maturity. The inverse ratio of production to biomass (P/B , see Appendix A) expresses size and longevity for each functional group, the unit of production and biomass is t/km^2 . (4) Optimising multiple criteria (i.e., economic, social and

ecosystem conservation) of the ecosystem is the final scenario. We set weighting factors for each criterion. For the economic criterion, for example, a weight of 1 to represents an emphasis on the economic net profit. For the social criterion, a weight of 1 indicates that the social objective has the same importance as the economic criterion. All three objectives are set at 1.0 in the compromise scenario. A weight of 0.01 used for the criterion under optimal conditions. The annual economic discount rate remains at 4% for all simulation.

In addition to the optimisation of the fishing effort of fleets, the effect of increased weights placed on profits, jobs/catch value and ecosystem structure on the average trophic level (TL) of catches, total catches and Kempton's index of diversity are examined. Kempton's index is calculated as $Q_{75} = S/[2 \log(B_{0.75s}/B_{0.25s})]$, where S is the number of functional groups and $B_{0.75s}$ and $B_{0.25s}$ are upper and lower 25 percentiles of the biomass values, respectively.

Because the results of the Ecosim simulation are sensitive to the vulnerability (V) setting, each simulations is repeated with three different vulnerabilities: the default vulnerability (2), the vulnerabilities estimated with the time series fitting, and the vulnerabilities that are proportional to the trophic level of the predators. In addition, two different discount rates are applied here to test the effect on societal benefits, 4% indicates a future generation friendly rate, and 15% means a higher rate presenting the fisheries resource under overexploited situation.

3. Results

3.1. Model construction

A previously constructed Ecopath model for PRE (Duan, 2009) is developed that comprises 34 functional groups with 15 commercially exploited species. The basic input for each group and the parameters that are used are presented in Appendix A. The selection of the groups is based on their ecological and economic importance. A box model describing the major flows of biomass is shown in Fig. 2.

Table 1

Cost related to the operation of different fleet on the PRE.

Fleet name	Fixed cost (%)	Effort related cost (%)	Sailing related cost (%)	Profit (%)
Pair and stern trawler	73.31	3.63	19.61	3.45
Shrimp trawler	60.00	5.00	20.00	15.00
Purse seine	42.39	26.29	20.92	10.40
Hook and Line	13.36	31.31	31.31	24.02
Gill net	6.18	27.28	27.28	39.26
Other miscellaneous	20.00	20.00	40.00	20.00

In Ecosim, the vulnerability factor is one of the most important parameters that determine the form of the predator–prey relationship. Therefore, this parameter was estimated by fitting the PRE ecological model to the time-series data of the relative capture rates and fishing effort from 1981 to 2005. We show the best fitting in Fig. 3. The total sum-of-square error (SS) was minimised from 64.67 to 33.63 after varying the vulnerability (V) factor.

The model shows a shift from a system dominated by large demersal species to one dominated by small pelagic species under the current exploitation level, with all of the commercially exploited species depleted. This suggests that the decrease in predator biomass will result in an increasing abundance of species, such as small pelagic species and benthic crustaceans, in the lower trophic levels.

3.2. Policy simulation

Maintaining the current exploitation levels produces a large risk for the fishing industry because of the decreasing biomass stock. EwE incorporates numerically optimal fishing levels that maximise the benefits under the state management objectives. After calibrating the Ecosim model, the fishing policies are explored by accounting for the maximisation of the objective functions for the ecological, economic and social criteria. Scenarios for the three independent criteria, as well as their interactive combinations are shown in Table 2. The resulting changes in fishing effort under several different scenarios are shown in Fig. 4.

In the single criterion scenario, such as the economic criterion scenario, we assign 1 to the weight factor ($W=1$), which indicates an emphasis on the net profits of fishing. The policy optimisation causes changes in the distribution of fishing effort among the fishing methods. We recognise that improving economic benefit cannot easily be accomplished as the level of effort increases. The results show that increasing the fishing effort of pair and stern trawling and shrimp trawling are at the expense of the efforts of purse seine, hook and line, gill-net and other miscellaneous methods. Because shrimp trawlers are the most profitable in the PRE, the effort is nearly 5.07 times the maximum gain of the net profit. Despite the well-documented damage on the benthic organisms (Hall, 1999; Hiddink et al., 2006), trawling activities are effective and provide an important food supply for humans. However, large increase in

this fishing effort will cause large changes in ecosystem structure. Although the overall biomass of the group is increased by 1.19%, most of the functional groups declined (or went extinct), particularly *C. lucidus*, *T. japonicus*, *D. maruadsi*, Trichiurids, small pelagic fish and benthopelagic fish (Fig. 5). The overall profit optimisation is predicted to lead to 183% growth in profits, which is associated with an 82% increase in jobs/catch value, a 92% increase in ecological structure, and a nearly 14% decrease in biodiversity.

To examine the influence of the social criterion, we assigned $W=1$; the simulation results suggested that there would be large benefits not only for the social criterion but also for net profits. This optimisation results in a 1.75-fold and 1.78 fold growth in net profits and in the jobs/catch value, respectively. Under this extreme scenario, the ecosystem structure is greatly altered, with a large loss of biodiversity. The total biomass of the functional groups is increased by 0.89%, while most commercially exploited species became extinct, including *C. lucidus*, *T. japonicus*, *D. maruadsi*, small pelagic fish, large pelagic fish and large demersal fish. The catch composition is dominated by “small fishes, shrimps and crabs”, at the expense of large fishes (Fig. 5).

Ecosystem optimisation leads to a reduction in the fishing effort of all of the fleets. In particular, net profits and employment decrease to 82% and 77% of the baseline level, respectively. The ecological structure optimisation results in an increase in the biomass of most species, notably crabs, *U. bensasi*, *N. virgatus* and large pelagic fish. This increase occurs concurrently with decrease in the biomass of small pelagic fish. This fishing strategy would lead to ecological reconstruction of food web, rather than decreasing the average trophic level of the catch (Pauly et al., 1998). The simulation for ecological criterion maximisation suggests a completely different policy from the economic and social optimisation processing, which would lead to a reduction in the fishing effort of almost all of the fleets, except for those using other miscellaneous methods (Fig. 4).

In the optimisation for a compromise solution all three objectives have a weight of 1. The results suggest that fishing effort increases by 144%, 151% and 4% for pair and stern trawling, shrimp trawling and hook and line fishery, respectively. The compromise solution increased the economic, social and ecological criteria by 138%, 56% and 93%, respectively. However, there was a decrease in biodiversity: *P. japonicus*, *T. japonicus*, *D. maruadsi*, Trichiurids,

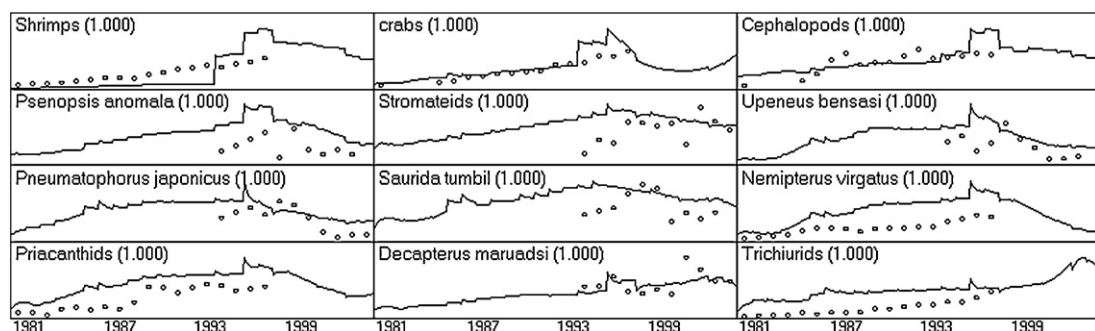
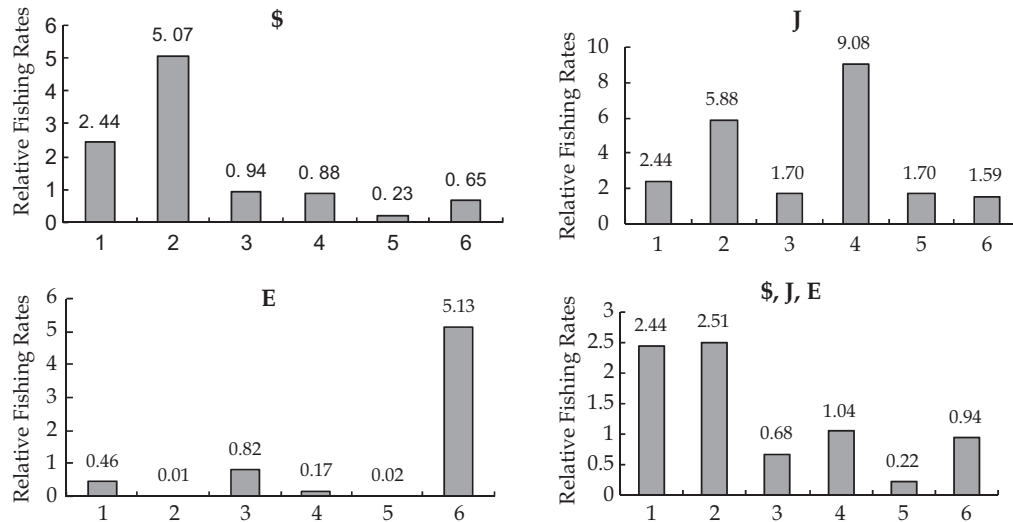


Fig. 3. Time series fit for commercially exploited function groups from 1981 to 2005.

Table 2

Results from optimisations for economic, social and ecological objectives and for all three objectives combined.

No.	\$	J	E	Total	Net economic value	Social value (employment)	Ecosystem structure	Diversity
S1	1	0.01	0.01	2.81	2.83	1.82	1.92	0.86
S2	0.01	1	0.01	2.77	2.75	2.78	1.64	0.76
S3	0.01	0.01	1	2.55	0.82	0.77	2.59	0.89
S4	1	1	1	1.96	2.38	1.56	1.93	1.02

**Fig. 4.** Related fishing mortality for fleets with different policy scenarios. Notes: (1) Pair and stern trawling, (2) Shrimp trawling, (3) Purse seine, (4) Hook and line, (5) Gill net, and (6) Other miscellaneous methods.

small pelagic fish, benthopelagic fish and large demersal fish all went extinct in this scenario, and there was a 2% increase in biodiversity.

Despite a decreasing trend of commercially exploited species under current conditions (Fig. 6), the Ecosim simulations suggest that the biomass stocks could recover under either the ecological objective maximisation or compromise scenarios. This indicates that successful management can be developed, that is based on fleets control and that accounts for the dynamics of trophic relationships.

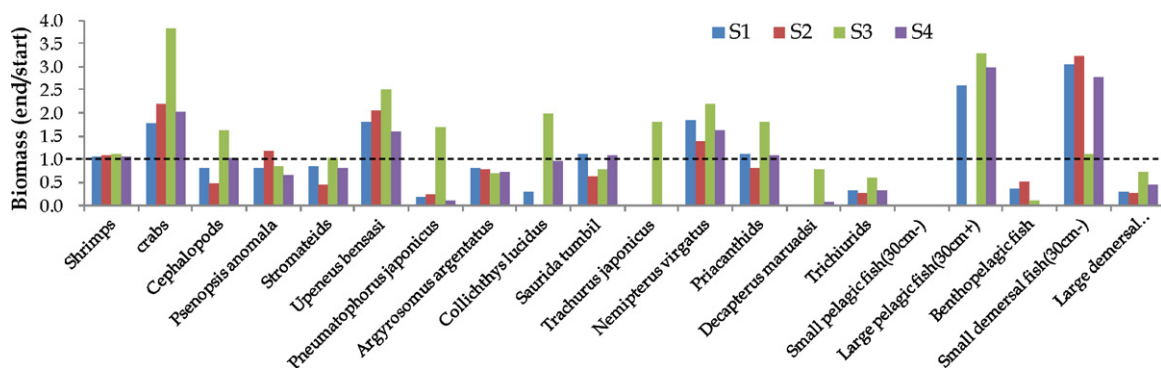
3.3. Effects of changing weights on the ecosystem indices

The effects of changing the relative weights of the net profit, employment and ecosystem structure objectives on the diversity, the trophic level of the catch and total catch are shown in Fig. 7. The weights for the two constant objectives are set to 0.2 in all of

the simulations, and the objective functions vary from 0 to 1, in increments of 0.1.

We use biomass diversity as one measure of the ecosystem structure. A relative index of biomass diversity is calculated with a modified version of Kempton's Q_{75} index which was originally developed for expressing species diversity. The average trophic level of the catch is another criterion for ecological structure. Previous studies showed that fisheries under heavy fishing strain result in a total catch increases, while the average trophic level of the catch decreases (Pauly et al., 1998).

As the weight on net profits increase, there was a steady upward trend in the total catch and a decline in the average trophic level of the catch. As expected, the effects of varying the weights that are placed on jobs/catch value shows similar trends as those for the maximisation of profits. In contrast, increasing the weights of the ecological objectives leads to increases in the average trophic level of the catch and small increases in Kempton's Q_{75} diversity, but an overall decrease in the total catch. The results of the optimisations

**Fig. 5.** Biomass ratios of the function groups from beginning of 30-year simulations to the end of the simulations for the 1998 Pearl River Delta. Notes: (S1) Optimising the economic benefits, (S2) Optimising the social benefits, (S3) Optimisation of the ecosystem status, (S4) Optimising multiple criteria.

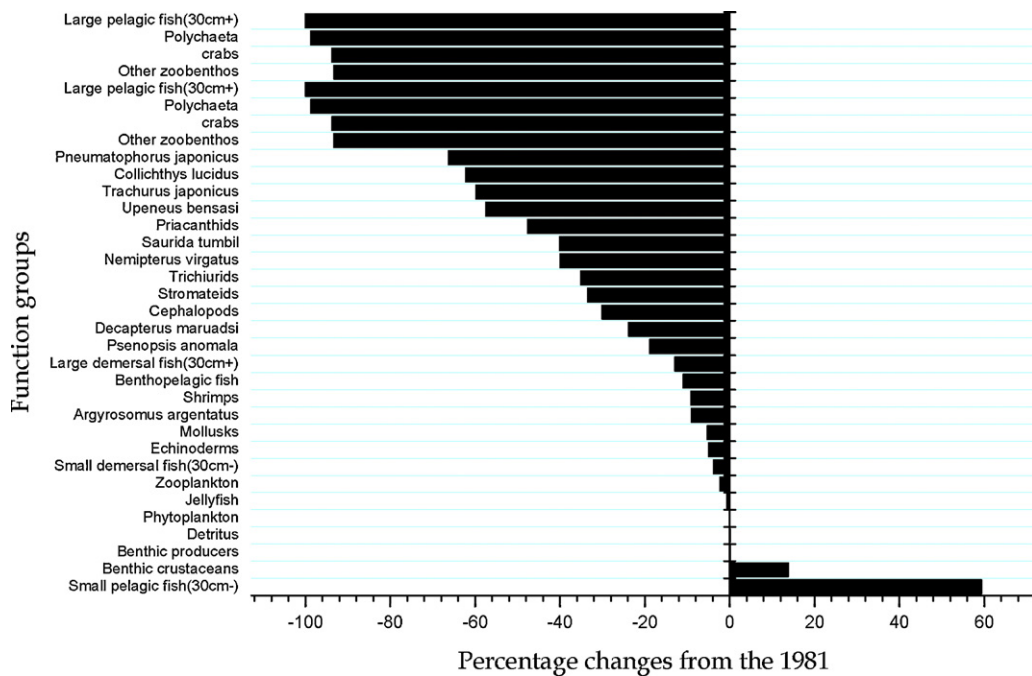


Fig. 6. Percentage changes of biomass from 1981 to 2015.

show that the average trophic level of the catch was quite resistant to the changes in the underlying system structure. There were no clear trends for the diversity index as the weight on profits is varied, and the trophic level of the catch demonstrated the opposite variation compare to the total catch. Similar results have been reported by (Araujo et al., 2008a). When all functions are given a weight of 0.2, the biodiversity index (Kempton's Q_{75}) is 2.64, the average trophic level of the catch is 2.95, and the total catch is 24.88 t/km².

3.4. Vulnerability

Vulnerability is a key parameter that is uncertain in the Ecosim model, and the results can be sensitive to the vulnerability

settings, particularly if the system is strongly under top-down control (Pauly and Christensen, 1993). A top-down controlled system can create large fluctuations in the biomass between prey and predator groups. In this study, the observed time-series catch and fishing effort data from the PRD in 1981–2005 are used to estimate the vulnerability factors, providing empirical support for the Ecosim model. When time-series data from the PRE are available in the future, the model can be further validated by comparing the observed data with the model predictions. Here, three different settings are compared, the default setting (2), the vulnerability estimated with the time-series fitting and the vulnerability that is proportional to the trophic level of the predators as in Fig. 8.

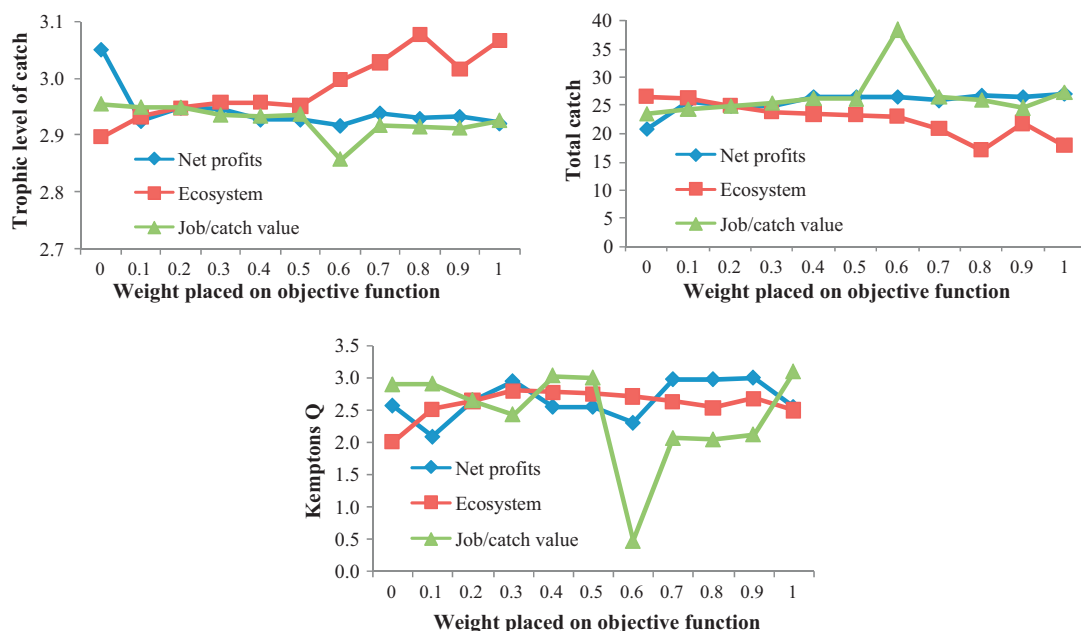


Fig. 7. Results of the weights placed on profit, jobs/catch value and the ecosystem structure.

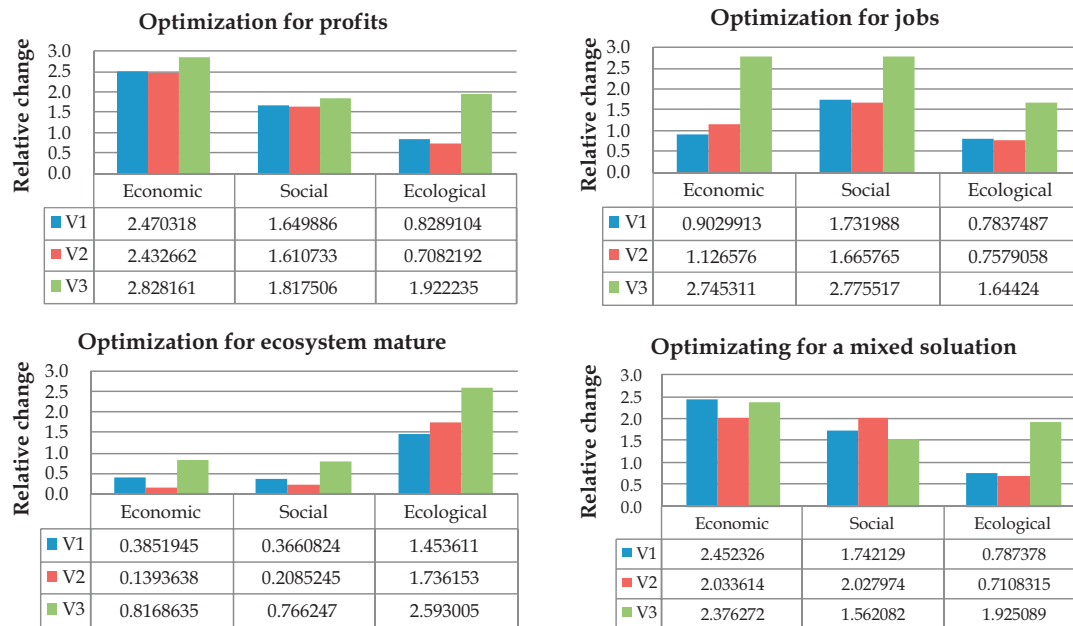


Fig. 8. Relative change in net profits, the jobs/catch value and the ecosystem structure that result from optimising for profits, jobs and the ecosystem with the three vulnerability parameters: (1) V1-Default, (2) V2-Vulnerability estimated with time-series fitting and (3) V3-Vulnerability proportional to the trophic level of the predators.

Similar to previous studies (Araujo et al., 2008b; Chen et al., 2008), the simulation resulted in the highest values of profits, jobs/catch value and ecosystem objectives when vulnerability was proportional to the trophic level of the predators. The most striking difference among the three settings occurred for the optimisation of jobs. However, there was a small discrepancy in the optimisation of the mixed solution, where the highest values for economic and social objectives were obtained under the V1-Default value 2 and V2-Vulnerability, respectively, which were estimated with time-series fitting. Over-all, using a vulnerability that was proportional to the trophic level of the predators appears to be an appropriate choice for the vulnerabilities settings in the Ecosim simulation.

3.5. Discount rate

Two different discount rates are applied to calculate the net present value. Usually, a lower discount means more weight is placed on future net benefits relative to the higher discount rates (Heymans et al., 2009). A discount rate within 2–5% will protect the future benefits to natural resources and intergenerational considerations (Weitzman, 2001; Sumaila and Walters, 2005). However, an annual discount rate within 10–20% would cause species extinction when the present harvest values from fisheries are maximised (Clark, 1973). In this study, we chose 4% as the rate that considers future generations, and 15% as the discount rate that is used when there are no regulated fisheries.

The results show that the discount rate is one of most important parameters when optimising for profits and job/catch value, although the effect differs in each scenario. Optimisation for profits with 4% discount increases profit more than 15% discount, whereas optimisation for jobs shows the opposite trend. The effects of the different discount rates on optimisation for ecosystem maturity are smaller than the other objectives, and the optimisation with lower discount rate obtains a higher value than that with a higher discount rate (Fig. 9).

4. Discussion

Managing an ecosystem for maximising integrated benefits creates a great challenge that is not easily achieved. In the case study of the PRE, the optimal fishing policies are explored with an ecosystem modelling approach. The management policy that is based on a multi-species model provides a clear interpretation of the trade-offs among economic, social and ecosystem objectives in fisheries management. All of the commercially exploited species are heavily harvested in the Ecopath model of 1998, and the stocks of some commercial exploited species are much lower than they were in 1980s (Duan et al., 2009b). This analysis indicates that the PRE fisheries in 1998 are sub-optimal in terms of achieving conservation and socio-economic objectives. Thus, the PRE fisheries can improve their conservation status by reducing fishing efforts of certain sectors. Previous studies have used Ecosim to explore fishing policy options for maximising the net economic value, job opportunities and ecological structure (Christensen and Walters, 2004b; Okey and Wright, 2004; Araujo et al., 2008b; Chen et al., 2008). The trade-offs among ecological, economic and social criteria are important problems for fisheries management, although an inverse relationship exists between ecological and socioeconomic goals. This makes it possible to find a balance when maximising the overall objects of the system (Zetina-Rejón et al., 2004).

Management policies have resulted in different degrees of recovery for the PRE ecosystem. We conclude that the performance of the fishing activities could be improved by adjusting the fishing effort of the fleets. The optimisation of a single objective leads to the specialisation of fishing fleets, with some fleets almost excluded from continuing fishing efforts. The optimisation for profits and jobs causes a loss of biodiversity, although the overall biomass of the functional groups is slightly increased. The optimisation for ecological structure, causes an increase in biodiversity and ecosystem maturity. Comparisons of the modelling optimisations indicate that restoring the depleted ecosystem would increase the total potential benefits (Table 2). The ideal solution places equal weight on profits, jobs and ecosystem status such that economic performance

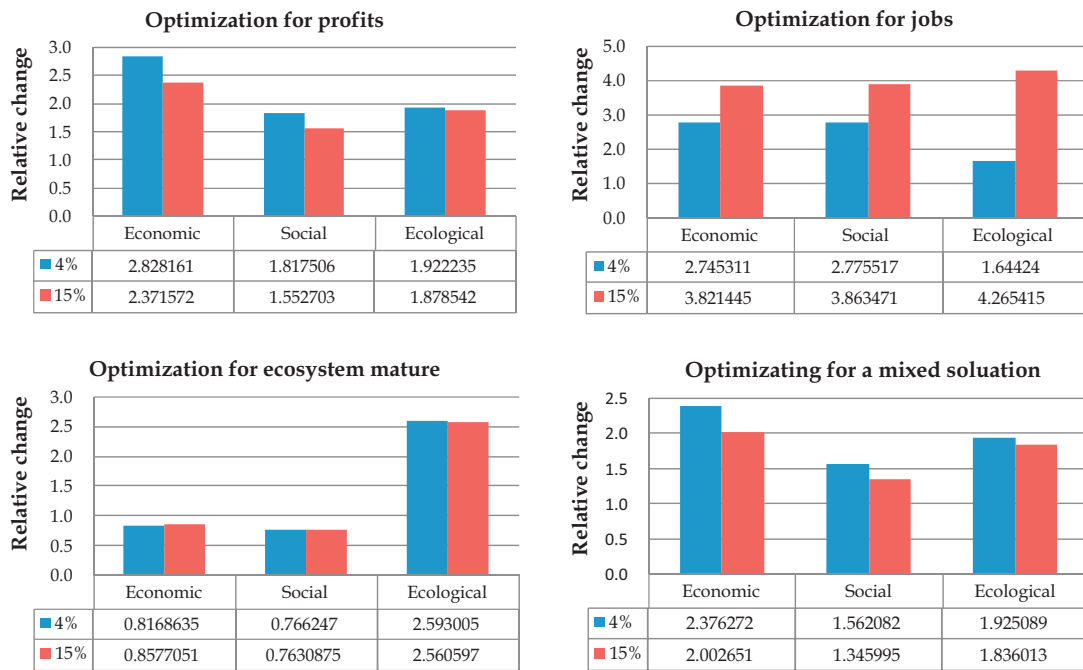


Fig. 9. Relative changes in the net profits, jobs/catch value and ecosystem structure that result from the optimisation for profits, jobs and ecosystem with the 4% and 15% discount rates.

will increase by 1.38 fold, social performance by almost 56%, and ecological performance by 93%. There is also 2% improvement in biodiversity.

We recognise that there is risk within the framework developed in this analysis. For example, the optimisation of economic goals is achieved by reducing top predators, and the whole system achieves higher benefits under this scenario. The patterns and mechanisms are similar to those in the social optimisation scenario. Ecosystem improvement reduces the catches of over-exploited and more productive species. As observed by (Cheung and Sumaila, 2008), shrimp trawling, stern and pair trawling and gill-netting are greatly reduced in the ecosystem optimisation scenarios. Because shrimp trawlers and gill net mainly target valuable invertebrates and fishes, reducing these two methods may have a negative effect on the net profits and employment opportunities from fisheries (Table 2). Moreover, maximise ecological stability results in decreasing both economic and social benefits. It is apparent that optimisation for the ecosystem does not create as many changes as optimisation for profits and jobs. This is because it is very difficult to change the turn-over rate of functional groups, and the optimisation procedures maximise the biomass/production (B/P) ratio of the species (Christensen et al., 2005).

Previous studies (Fulton and Smith, 2004; Araujo et al., 2008b; Cheung and Sumaila, 2008) suggested that Ecosim simulation results are very sensitive to the vulnerabilities parameters. Three different settings for vulnerabilities were analysed in this paper. The results indicate that the vulnerabilities that are proportional to the trophic level of the predators are better than the default (2) vulnerabilities and when the vulnerability was estimated with time series fitting. Hence, we suggest that vulnerabilities proportional to the trophic level will be appropriate when running the Ecosim model.

For a renewable resource, such as fisheries, the present value of future benefits compared to the current benefits, primarily depends on the relationships between biological growth and the discount rate. In reality, as long as the population growth rate is greater than discount rate, extinctions will not occur. Otherwise, if the population growth rate is less than the discount rate, and the marginal

costs of fishing are low enough, species depletion would occur. In this paper, we compare the effect on whole fish communities with a lower discount rate (4%) and higher discount rate (15%). The results indicate that higher benefits are obtained from economic, social and mixed objectives optimisation. This suggests that fisheries should aim towards short-term economic benefits at the expense of long-term economic and ecological gains. However, maximising for jobs/catch shows an opposite trend that differs from previous studies (Heymans et al., 2009); it shows long-term benefits of profits and jobs/catch value can be obtained with a lower discount rate. The overexploitation of resources using a higher discount rate is a serious problem in environmental and resource management. However, if we use a lower discount rate, the economic profits and societal benefits can be increased. Hence, it is important to consider the benefit for future generation when exploring renewable resource.

To solve the problem of explosive growth in the number of fishing vessels, China's government launched an 8-year fishing vessel buyback program from 2003 to 2010. The aim of this fishing policy was to transfer surplus labour force to other fishing sectors or industries. Details on the size of the fishing boats and their fishing methods are not yet available. This study indicates that if fishing vessel buyback programs can be implemented under ecological optimisation or a compromise solution, the buyback policy may benefit fishery conservation and can improve societal benefits from fisheries.

5. Conclusion

It is widely recognised that fishing policy simulations can predict, or at least provide warnings of, unknown responses to fishing activities or fishery policy implementation. The aim of this study was to quantify the potential trade-offs between ecological and socio-economic objectives. The PRE has been greatly depleted by fishing and is considered to be in a sub-optimal state with respect to both ecological and economic objectives. In the southwestern region of the Gulf of Mexico, the impact of a specific shrimp-recovery strategy was evaluated (Arreguin-Sanchez et al., 2004,

2008). An analysis based on a single-species model did not offer clear expectation for the recovery of the shrimp stocks. In the best case scenario, assuming a stable (current) stock, and if the fisheries were closed for 10 years, the shrimp stock could recover by almost 20% (Ramírez-Rodríguez et al., 2000). We believe that the findings in this paper should be generally applicable to other tropical marine ecosystems where overfishing is commonly observed.

Understanding the trade-offs resulting from the different management objectives may allow fishermen and the public to have informed discussions on future management policies. It is critical to develop alternative livelihoods for fishermen who may be affected by fisheries management and conservation policies to facilitate the transition to sustainable fishing. The results obtained from Ecosim models for policy exploration are useful for providing a new approach for fisheries management that considers entire

fishing communities and cannot be obtained from studies on single species. The fishing fleets urgently need to be restructured to prevent ecosystem degradation and stock extirpation as well as to improve economic benefits from harvesting.

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Appendix A. Basic parameters of the PRD coastal ecosystem model (1998)

No.	Group name	Trophic level	B (t/km ²)	P/B (year ⁻¹)	Q/B (year ⁻¹)	EE	P/Q
1	Benthic producers	1.0	153	11.885	–	0.01	–
2	Phytoplankton	1.0	13	231	–	0.47	–
3	Zooplankton	2.0	10.4	32	192.00	0.31	0.167
4	Jellyfish	3.1	1.07467	5.011	25.04	0.74	0.200
5	Polychaeta	2.0	0.8	6.75	22.50	0.89	0.300
6	Mollusks	2.2	0.7	3.5	11.70	0.86	0.299
7	Echinoderms	2.3	0.24	1.2	3.58	0.88	0.335
8	Benthic crustaceans	2.2	0.56	5.65	26.90	0.84	0.210
9	Other zoobenthos	2.6	1.69	1	9.00	0.74	0.111
10	Shrimps	2.3		3.08	16.35	0.95	0.188
11	Crabs	2.5		3.79	12.50	0.95	0.303
12	Cephalopods	3.2	1.4747	3.1	8.00	0.95	0.388
13	<i>Psenopsis anomala</i>	3.0	0.1006	2.41	24.00	0.97	0.100
14	<i>Stromateids</i>	3.4	0.3930	3.03	15.15	0.93	0.200
15	<i>Upeneus bensasi</i>	3.1		2.0976	10.28	0.95	0.204
16	<i>Pneumatophorus japonicus</i>	2.8	0.0446	2.625	8.80	0.97	0.298
17	<i>Argyrosomus argentatus</i>	3.4	0.0341	3.55	7.71	0.96	0.460
18	<i>Collichthys lucidus</i>	3.3	0.0600	7.36	29.16	0.96	0.252
19	<i>Saurida tumbil</i>	3.3	0.0200	4.26	7.12	0.93	0.598
20	<i>Trachurus japonicus</i>	3.5	0.4120	2.15	7.86	0.90	0.274
21	<i>Nemipterus virgatus</i>	3.1	0.4860	2.07	7.25	0.93	0.286
22	<i>Priacanthids</i>	3.4	0.2160	2.94	8.00	0.92	0.368
23	<i>Decapterus maruadsi</i>	3.1	0.4607	1.87	11.08	0.92	0.169
24	<i>Trichiurids</i>	3.8	1.1997	3.02	6.21	0.91	0.487
25	Small pelagic fish (30 cm ⁻)	2.8	1.772	4.26	17.04	0.97	0.250
26	Large pelagic fish (30 cm ⁺)	3.1	0.368	4.26	6.27	0.96	0.679
27	Benthopelagic fish	2.8	0.922	3.08	15.42	0.91	0.200
28	Small demersal fish (30 cm ⁻)	2.6		4.7	23.50	0.95	0.200
29	Large demersal fish (30 cm ⁺)	3.0	0.1640	3.5	6.21	0.94	0.564
30	Sharks	3.8	0.0050	0.2	4.13	0.10	0.048
31	Seabirds	3.4	0.0030	0.06	66.10	0.06	0.001
32	Turtles	2.9	0.0002	0.1	2.50	0.10	0.040
33	Marine mammals	4.0	0.0090	0.045	14.77	0.05	0.003
34	Detritus	1.0	200	–	0.00	0.16	

B, biomass; P/B, production to biomass ratio; Q/B, consumption to biomass ratio; P/Q, production to consumption ratio; EE, ecotrophic efficiency.

Appendix B. Estimated fishery catch by function groups and fishing sectors in the 1998 model

	Group name	Pst	ST	PS	H&L	GN	Others	Total
1	Benthic producers	0.000	0.000	0.000	0.000	0.000	0.056	0.056
2	Phytoplankton	0.000	0.000	0.000	0.000	0.000	0.000	0.000
3	Zooplankton	0.000	0.000	0.008	0.000	0.000	0.102	0.110
4	Jellyfish	0.009	0.000	0.022	0.000	0.000	0.000	0.031
5	Polychaeta	0.000	0.000	0.000	0.000	0.000	0.000	0.000
6	Mollusks	0.020	0.047	0.000	0.000	0.000	0.132	0.199
7	Echinoderms	0.000	0.000	0.000	0.000	0.000	0.000	0.000
8	Benthic crustaceans	0.000	0.015	0.000	0.000	0.000	0.000	0.015
9	Other zoobenthos	0.000	0.002	0.000	0.000	0.000	0.000	0.002
10	Shrimps	0.080	0.576	0.000	0.000	0.000	0.023	0.679
11	Crabs	0.040	0.100	0.000	0.000	0.042	0.018	0.200
12	Cephalopods	0.332	0.053	0.146	0.002	0.000	0.070	0.603
13	<i>Psenopsis anomala</i>	0.041	0.003	0.007	0.000	0.007	0.014	0.072
14	<i>Stromateids</i>	0.294	0.028	0.056	0.109	0.307	0.076	0.870
15	<i>Upeneus bensasi</i>	0.021	0.000	0.000	0.000	0.000	0.000	0.021
16	<i>Pneumatophorus japonicus</i>	0.101	0.000	0.000	0.000	0.000	0.000	0.101
17	<i>Argyrosomus argentatus</i>	0.031	0.000	0.000	0.001	0.021	0.001	0.053
18	<i>Collichthys lucidus</i>	0.048	0.028	0.107	0.017	0.081	0.000	0.281
19	<i>Saurida tumbil</i>	0.004	0.005	0.000	0.000	0.052	0.013	0.073
20	<i>Trachurus japonicus</i>	0.169	0.002	0.281	0.000	0.000	0.000	0.452
21	<i>Nemipterus virgatus</i>	0.472	0.009	0.000	0.067	0.102	0.025	0.676
22	<i>Priacanthids</i>	0.279	0.006	0.000	0.041	0.063	0.016	0.404
23	<i>Decapterus maruadsi</i>	0.385	0.000	0.085	0.000	0.000	0.000	0.470
24	<i>Trichiurids</i>	2.272	0.045	0.018	0.075	0.593	0.153	3.155
25	Small pelagic fish (30 cm ⁻)	0.512	0.000	1.086	0.000	0.103	0.644	2.345
26	Large pelagic fish (30 cm ⁺)	0.227	0.000	0.090	0.017	0.408	0.079	0.821
27	Benthopelagic fish	0.342	0.028	0.061	0.001	0.058	0.113	0.603
28	Small demersal fish (30 cm ⁻)	0.009	0.076	0.010	0.006	0.026	0.053	0.179
29	Large demersal fish (30 cm ⁺)	0.052	0.207	0.010	0.002	0.065	0.016	0.352
30	Sharks	0.000	0.000	0.000	0.000	0.000	0.000	0.000
31	Seabirds	0.000	0.000	0.000	0.000	0.000	0.000	0.000
32	Turtles	0.000	0.000	0.000	0.000	0.000	0.000	0.000
33	Marine mammals	0.000	0.000	0.000	0.000	0.000	0.000	0.000
34	Detritus	0.000	0.000	0.000	0.000	0.000	0.000	0.000
35	Sum	5.737	1.229	1.986	0.337	1.928	1.602	12.819

Note: Pst, Pair and stern trawl; ST, Shrimp trawl; PS, Purse seine; H&L, Hook and Line; GH, Gill net.

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